1.1.

$$pV = nRT$$

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p = \frac{1}{T}$$

$$\beta = \frac{1}{p}\left(\frac{\partial p}{\partial T}\right)_V = \frac{1}{T}$$

$$k_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T = \frac{1}{p}.$$

1.2. (1). $V = V(T, p)$.

$$\int \frac{1}{V}\,dV = \int \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p dT + \frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T dp = \int \alpha\,dT - k_T\,dp.$$

$$= \int \frac{1}{T}\,dT - \int \frac{1}{p}\,dp \iff \ln V = \ln T - \ln p \iff pV = CT.$$

1.5.

$$dL = \left(\frac{\partial L}{\partial J}\right)_J$$